

Environmental Engg.

BRANCH: CIVIL ENGG. (CE)

COURSE CODE: CE401

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Environmental Engg. (CE401) under the Civil Engg. (CE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Water Demand & Quality

Water requirements, population forecasting methods (arithmetic, geometric, incremental increase). Physical, chemical, biological water quality parameters.

Unit 2: Water Treatment Systems

Sedimentation, coagulation, filtration (slow sand, rapid sand), disinfection (chlorination, ozonation). Aeration, softening methods.

Unit 3: Waste Water Characteristics

Sewerage systems, dry weather flow. Wastewater analysis: BOD, COD, suspended solids, pH. Biochemical Oxygen Demand kinetics.

Unit 4: Sewage Treatment & Disposal

Primary treatment: screening, grit chambers. Secondary treatment: Activated Sludge Process (ASP), Trickling Filters. Sludge digestion, septic tanks, disposal methods.

*BOD remaining at time t: $L_t = L_0 * e^{-(k*t)}$. Population projection (Geom): $P = P_0 * (1 + r)^n$.*

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



